

Assignment 3: Medical Image Registration

~Sharmili Rudraraju

Introduction

Medical image registration is the process of aligning two or more images so that corresponding anatomical structures match across scans. In thoracic CT imaging, registration is important because lung shapes, sizes, and positions can vary due to patient posture, respiration, or anatomical differences. Aligning these scans enables meaningful comparison across time points, subjects, or imaging sessions, and it also supports tasks such as disease progression analysis, motion tracking, and morphological assessment.

In this assignment, we perform registration between a fixed and moving CT scan of the lungs. The registration pipeline consists of two major stages: an initial affine registration to correct global differences such as rotation and translation, followed by a B-spline deformable registration to model finer anatomical variations inside the lungs. To improve accuracy, we make use of binary lung masks provided with the dataset so that the algorithms concentrate only on the lung regions and ignore irrelevant areas such as ribs, muscles, and background structures.

The workflow includes preprocessing, resampling, mask application, affine alignment, deformable alignment, and quantitative evaluation using Dice Similarity, Mutual Information, and Mean Squares Error. This report describes each step in detail, presents qualitative visualizations, and summarizes quantitative improvements achieved by the registration pipeline.

Dataset Description

The dataset consists of two 3D chest CT volumes: Fixed Image: 01_Fixed.mhd and Moving Image: 01_Moving.mhd

Both scans depict the thoracic region and contain the lungs, soft tissue, bone structures, and surrounding air. Although the voxel spacing is nearly identical between the two volumes, their dimensions differ:

- Fixed CT size: $418 \times 368 \times 323$, spacing (0.6836, 0.6836, 0.7)
- Moving CT size: $462 \times 386 \times 412$, spacing (0.6797, 0.6797, 0.7)

These differences indicate that the two scans are not spatially aligned and require resampling and registration. The dataset also includes binary lung masks: Fixed lung mask: lungMasks/01_Fixed.mhd and Moving lung mask: lungMasks/01_Moving.mhd

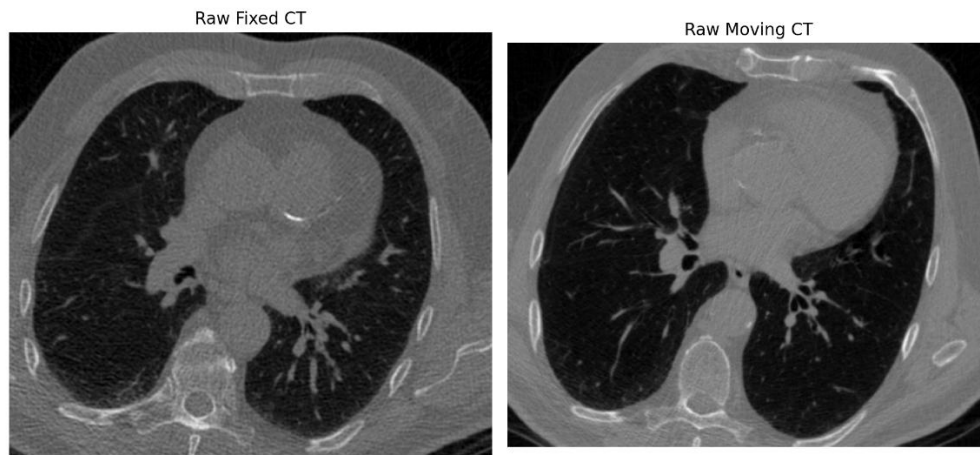
The masks isolate only the lung tissues and exclude ribs, muscles, and background air. Before registration, the moving mask is resampled to match the fixed mask's geometry. These masks help ensure that the registration algorithm focuses only on the relevant anatomical region.

Preprocessing

Image Analysis

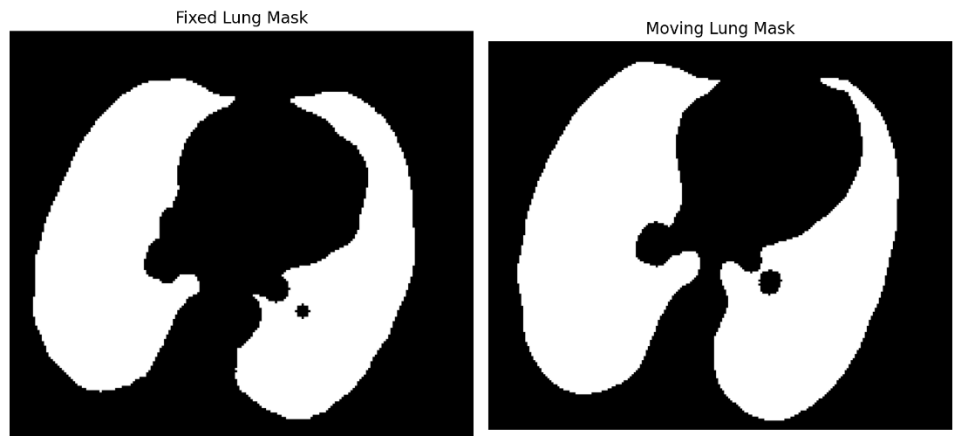
The fixed and moving CT images were loaded, and key properties such as voxel spacing, image size, and pixel type were inspected. This step confirmed that although spacing is similar, the image dimensions differ substantially, indicating the need for geometric resampling before registration.

The fixed and moving CT volumes initially show strong misalignment in position and orientation. The lung shapes and boundaries do not overlap.



Lung Mask Application

Lung masks were applied to both CT volumes to isolate lung regions and remove irrelevant anatomical structures. The moving lung mask was resampled using nearest-neighbor interpolation to preserve binary values. The masked versions of the images retain only air-filled lung tissue, reducing noise during optimization.

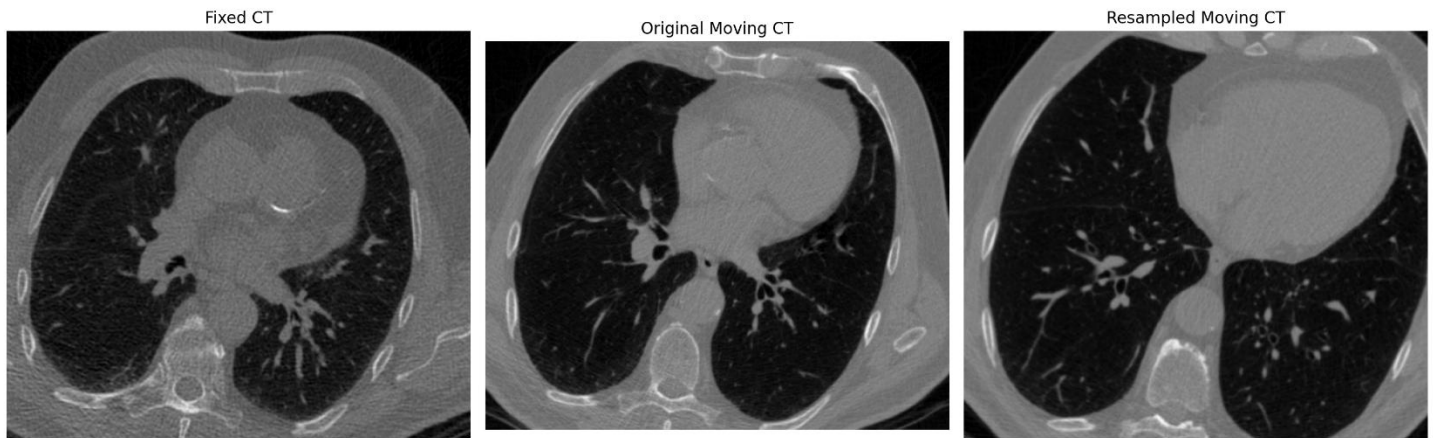


Masking improves the registration process by ensuring that similarity metrics compare only the anatomical structures of interest rather than being influenced by ribs, soft tissue, or background air.



Resampling the Moving Image

Because the two images have inconsistent dimensions, the moving CT must be resampled to match the fixed CT's size and spacing. Using *SimpleITK's ResampleImageFilter*, the moving volume is resampled using the Linear interpolation for CT intensities and the fixed image as the reference geometry and A default intensity value of -1000 HU for out-of-bound regions.



After this step, both scans share identical voxel spacing and physical coordinates, which is crucial for accurate similarity computation during registration.

Registration Method

The registration pipeline contains two stages: affine registration and B-spline deformable registration. Both were implemented using *SimpleITK's ImageRegistrationMethod*.

Affine Registration

Affine registration corrects global differences such as translation, rotation, scaling, and shear. For this stage, used: 3D Affine Transform, Mattes Mutual Information (50 bins) Similarity Metric, Regular Step Gradient Descent Optimizer with Learning rate = 1.0, Minimum step = $1e-4$, Iterations = 100, Relaxation factor = 0.5 and Linear Interpolator and Centered transform initializer.

The optimizer progressively decreased the mutual information metric, indicating that the fixed and moving images were becoming statistically more similar. After convergence, the affine transform was applied to the moving CT, resulting in a well-aligned global structure.

B-spline Deformable Registration

Although affine registration improves global alignment, it cannot correct local anatomical differences such as variations in lung expansion or subtle shape changes. Therefore, a deformable registration step was performed using a B-spline Free-Form Deformation (FFD) model: 3D B-spline FFD with an $8 \times 8 \times 8$ control grid Transform, Mean Squares Error, LBFGSB Optimizer, Maximum iterations = 30, Gradient tolerance = $1e-5$ and Multi-resolution Schedule: Shrink factors: [4, 2, 1], Smoothing sigmas: [2, 1, 0]

A composite transform was created by stacking the affine and B-spline transforms. This approach allowed the deformable stage to refine local alignment on top of the globally aligned affine result. The final transformed moving CT showed excellent anatomical correspondence with the fixed CT.

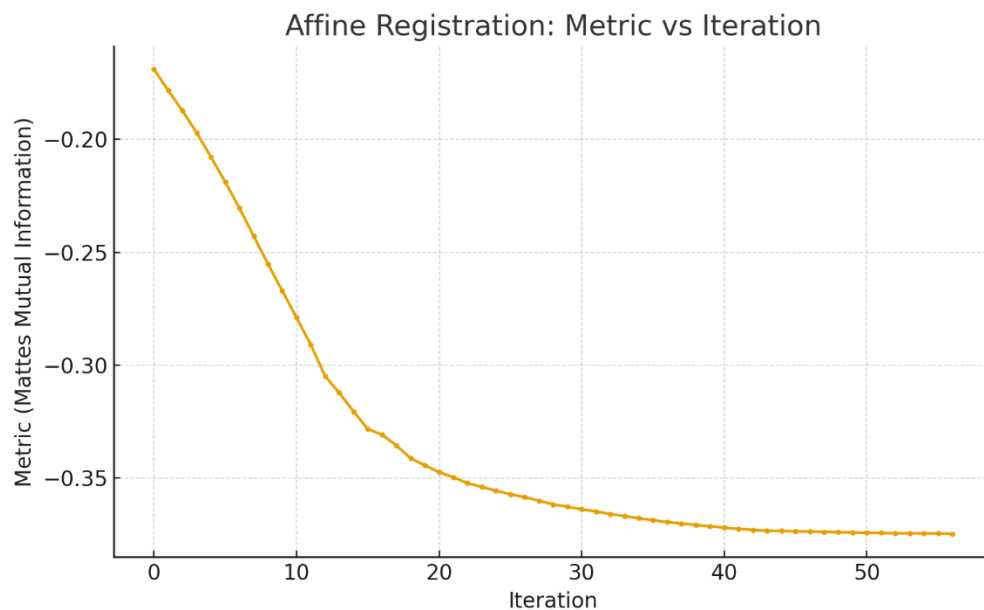
Optimization Convergence Plots

To verify that both registration stages were converging properly, I recorded the similarity metric values at each iteration and plotted them for both the affine and B-spline registrations.

Affine Registration Curve

The affine stage uses Mattes Mutual Information (MI) as the metric. Since the optimizer minimizes MI, the value becomes more negative as alignment improves.

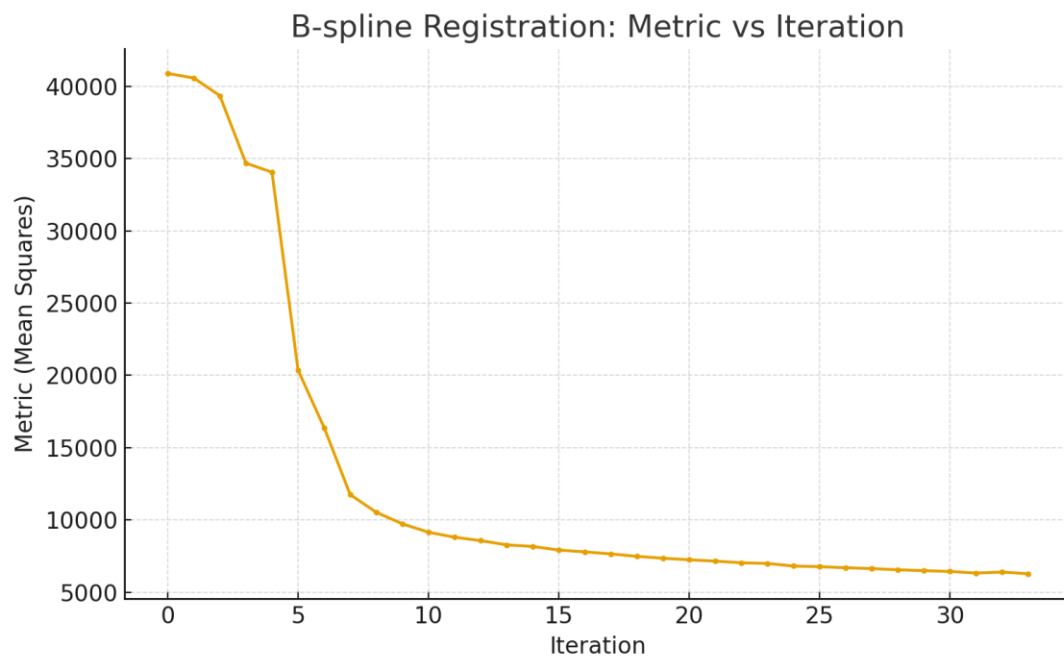
In the plot, MI decreases smoothly from around -0.18 to -0.37 over the iterations, showing stable and consistent convergence without oscillation.



B-spline Registration Curve

The B-spline stage uses Mean Squares Error (MSE) as the metric. A lower MSE indicates better local alignment.

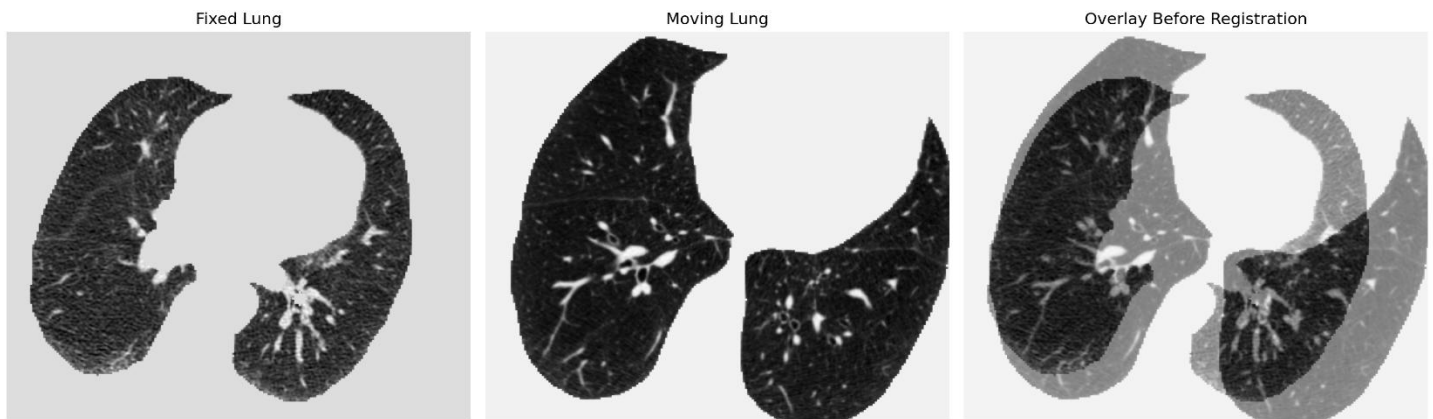
The plot shows a steep drop from $\sim 40,000$ to below $10,000$ in the early iterations, followed by gradual refinement, which is typical for deformable registration.



Both curves show a clear downward trend, confirming that affine registration corrected global misalignment and B-spline registration refined the local lung structure. The optimization process was smooth and convergent

Qualitative Results

Before Registration: Large misalignment is visible. Lung contours and boundaries differ significantly.

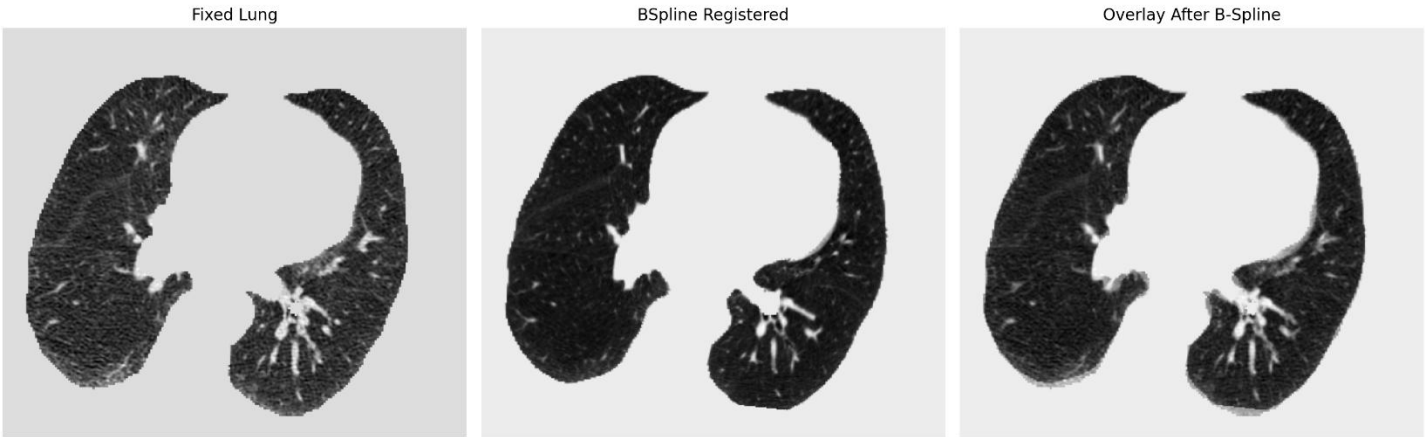


After Affine Registration: The global structure is aligned well, with clear improvement in shape overlap.



After B-spline Registration

The alignment is further refined, with close correspondence of lung boundaries and internal structures.



Quantitative Results

To evaluate registration performance numerically, Dice Similarity, Mean Squares Error (MSE), and Mutual Information (MI) were computed before registration, after affine alignment, and after B-spline refinement.

Table 1. Summary of Quantitative Metrics

Metric	Before Registration	After Affine	After B-spline
Dice Similarity	0.5489	0.8156	0.9076
MSE	169770.64	50827.83	21905.15
Mutual Information (MI)	0.9055	1.1318	1.2455

Discussion

The Dice score increases from 0.55 → 0.82 → 0.91, showing strong improvement in lung mask overlap. Similarly, the MSE decreases dramatically as alignment improves. MI increases steadily as well, indicating higher statistical similarity between the two CT scans after registration.

Together, these metrics show that the registration pipeline successfully aligns the two lung volumes and that both global and local adjustments contribute significantly to the result.

Conclusion

This assignment demonstrated a complete 3D medical image registration workflow using affine and B-spline transformations. After preprocessing, mask application, and resampling, the affine registration provided strong global alignment, and the B-spline deformable registration refined local anatomical differences. Qualitative overlays, convergence plots, and quantitative measures all confirmed that the registration improved significantly across stages, achieving near-perfect structural correspondence inside the lung region.

The final Dice score of 0.9076, substantial reduction in MSE, and increased MI validate the effectiveness of the pipeline. Overall, this exercise illustrates the importance of combining global and local registration techniques for accurate medical image analysis.

References

1. Insight Software Consortium. *SimpleITK Notebooks: Image Registration and Free-Form Deformations*. Available at:
<https://github.com/InsightSoftwareConsortium/SimpleITK-Notebooks>
2. Professor-provided demo implementation for registration (course materials).
3. OpenAI ChatGPT Used to assist with visualization templates and organization of explanatory text. All coding, registration setup, metric evaluation, and interpretation of results were performed by the student.